Advanced Reading: MoBA and Learned Block-Sparse Attention

Deep Dive into Kimi K3 — Chapter 11

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Foundation Books Project

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Goal: type block routing against dense attention, prove causality and the dense limit, execute it exactly with route identity and online-softmax merges, count its real cost, and read the reported evidence in scope. MoBA is a separate study, not a documented K3 layer.

Roadmap

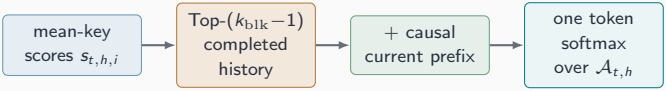
Selection and causality

Exact execution

Cost, transitions, and evidence

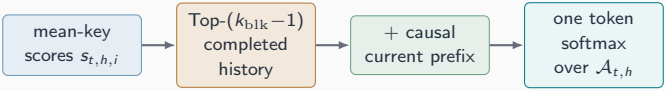
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- $\mathcal{A}_{t,h} \subseteq C_t$, and selecting every completed history recovers dense causal attention. Trace: Head 1 at τ_6 reads $\{0, 1, 2, 3, 8, 9\}$.

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- Partitions return (m, ℓ, u) and merge exactly: $27/7$, not the averaged $11/3$.
- Partitioned forward matches the dense masked oracle to 4.45×10^{-16}. [OBSERVED: book companion code, CPU]

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Diagnose from the earliest violated contract. Next: Chapter 12, Scaling Muon for LLM Training.